

MOPTI, Mali

WMO: 612650

Lat: 14.513N

Lon: 4.080W

Elev: 906

StdP: 14.22

Time Zone: 0.00 (E00)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	58.0	60.4	17.3	13.6	87.6	17.6	13.8	87.9	21.1	82.2	19.1	81.3	3.5	40	0.433

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
5	21.5	108.0	69.1	106.6	69.2	105.0	69.2	80.7	88.9	79.9	88.6	79.2	88.3	8.1	40

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
78.9	155.0	84.3	77.9	150.0	84.0	77.1	145.8	83.8	45.1	88.7	44.3	88.7	43.5	88.4	84.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
20.9	18.1	16.0	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
			43.6	82.6	1.1	1.6	42.9	83.7	42.2	84.7	41.6	85.5	40.8	86.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	84.9	74.8	80.7	87.2	92.3	94.2	91.2	85.8	82.6	84.0	86.1	82.4	76.8
	DBStd	6.81	4.21	5.24	5.05	3.41	3.18	3.97	3.77	2.78	2.67	2.40	2.86	4.02
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	1	1	0	0	0	0	0	0	0	0	0	0	0
	CDD50	12723	768	859	1154	1269	1371	1235	1111	1011	1021	1120	972	832
	CDD65	7248	303	439	689	819	906	785	646	546	571	655	522	367
	CDH74	99909	3182	5429	10239	13338	15085	12223	8574	6097	6850	8754	6288	3850
CDH80	59332	1416	3045	6554	9188	10660	8020	4575	2549	3131	4765	3552	1877	
Wind	WSAvg	7.0	6.9	7.7	7.4	7.1	8.0	9.2	8.0	6.3	5.4	5.2	5.9	7.1
Precipitation	PrecAvg	19.4	0.0	0.0	0.0	0.2	0.6	2.6	5.4	6.8	3.3	0.8	0.0	0.0
	PrecMax	33.5	1.2	0.0	0.6	1.7	2.7	8.3	8.3	10.7	9.0	2.4	0.4	0.0
	PrecMin	10.4	0.0	0.0	0.0	0.0	0.0	0.6	2.7	2.8	0.7	0.0	0.0	0.0
	PrecStd	5.3	0.2	0.0	0.1	0.4	0.7	1.9	1.4	2.1	1.9	0.8	0.1	0.0
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	97.0	103.1	107.2	109.7	110.1	108.2	102.0	95.9	97.2	102.1	100.6	97.5
		MCWB	61.6	63.7	65.4	67.3	70.4	72.0	75.4	77.8	76.6	67.3	64.9	62.9
	2%	DB	92.7	100.3	105.0	107.9	108.2	105.8	99.3	93.4	94.7	99.6	98.7	95.2
		MCWB	60.1	63.0	65.2	67.3	70.2	74.1	75.8	77.9	78.1	69.1	64.0	61.6
	5%	DB	90.2	98.0	103.2	106.3	106.8	103.6	97.0	91.6	92.9	97.3	97.1	92.9
		MCWB	58.9	62.2	64.9	66.9	70.5	74.6	75.7	77.9	78.3	71.4	63.9	61.0
	10%	DB	87.7	95.1	100.9	104.5	105.2	101.4	94.7	89.9	91.3	95.2	95.0	89.8
		MCWB	57.9	60.8	63.9	67.0	70.9	74.5	76.0	77.7	78.1	73.4	63.7	60.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	65.1	68.4	73.3	76.7	78.3	79.4	80.2	81.0	81.8	81.8	76.5	68.3
		MCDB	86.5	94.4	97.1	98.3	96.5	96.2	89.4	88.9	89.6	88.7	86.1	90.4
	2%	WB	62.4	65.8	71.1	75.1	77.2	78.2	79.2	79.9	80.8	80.1	74.1	65.6
		MCDB	86.0	92.2	93.7	95.0	97.1	95.3	89.6	88.1	88.9	88.4	85.2	85.7
	5%	WB	60.8	64.2	68.9	73.9	76.3	77.4	78.4	79.2	80.1	79.3	71.3	63.6
		MCDB	85.9	92.4	93.6	94.5	96.7	94.1	89.3	87.3	88.2	88.2	83.4	85.3
	10%	WB	59.2	62.3	66.6	72.7	75.4	76.7	77.8	78.6	79.4	78.4	68.8	61.9
		MCDB	83.9	91.1	94.3	94.2	96.2	93.3	88.6	86.6	87.6	87.8	84.4	84.3
Mean Daily Temperature Range	5% DB	MDBR	26.2	26.4	25.8	23.9	21.5	18.7	16.0	13.7	14.8	19.2	26.1	26.6
		MCDBR	29.4	29.1	27.6	26.5	23.8	21.0	18.4	16.0	16.8	23.3	28.4	29.1
		MCWBR	10.3	9.9	9.0	9.0	7.6	5.7	4.4	5.1	6.1	7.9	9.2	9.9
	5% WB	MCDBR	28.3	26.6	22.3	20.4	20.6	18.9	16.0	14.5	15.5	17.0	23.7	27.8
MCWBR		10.5	9.7	9.0	8.3	7.0	5.9	5.4	5.7	6.4	7.2	9.7	10.9	
Clear-Sky Solar Irradiance	taub	0.526	0.558	0.679	0.701	0.725	0.714	0.638	0.588	0.573	0.589	0.528	0.501	
	taud	1.848	1.756	1.532	1.561	1.530	1.590	1.762	1.873	1.894	1.826	1.859	1.907	
	Ebn at Noon	243	242	218	213	205	204	221	235	239	231	240	246	
	Edh at Noon	63	73	93	90	91	85	72	65	64	67	62	58	
All-Sky Solar Radiation	RadAvg	1705	1901	2047	2104	2052	1928	1906	1846	1942	1911	1785	1673	
	RadStd	102	118	109	113	82	56	68	89	76	45	50	63	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (1 neighbor)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page